

Lab 1

Switch <-> LED Interface

By: Ben Robles & Robert Stevenson
Group U

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Verilog:

```
`timescale 1ns / 1ps

module switch_led_interface(
    input  wire [15:0] sw,
    output wire [15:0] led
);

    assign led = sw;

endmodule
```

XDC Snippet:

```
## Switches
set_property PACKAGE_PIN J15 [get_ports {sw[0]}]
set_property IOSTANDARD LVCMOS33 [get_ports {sw[0]}]

...

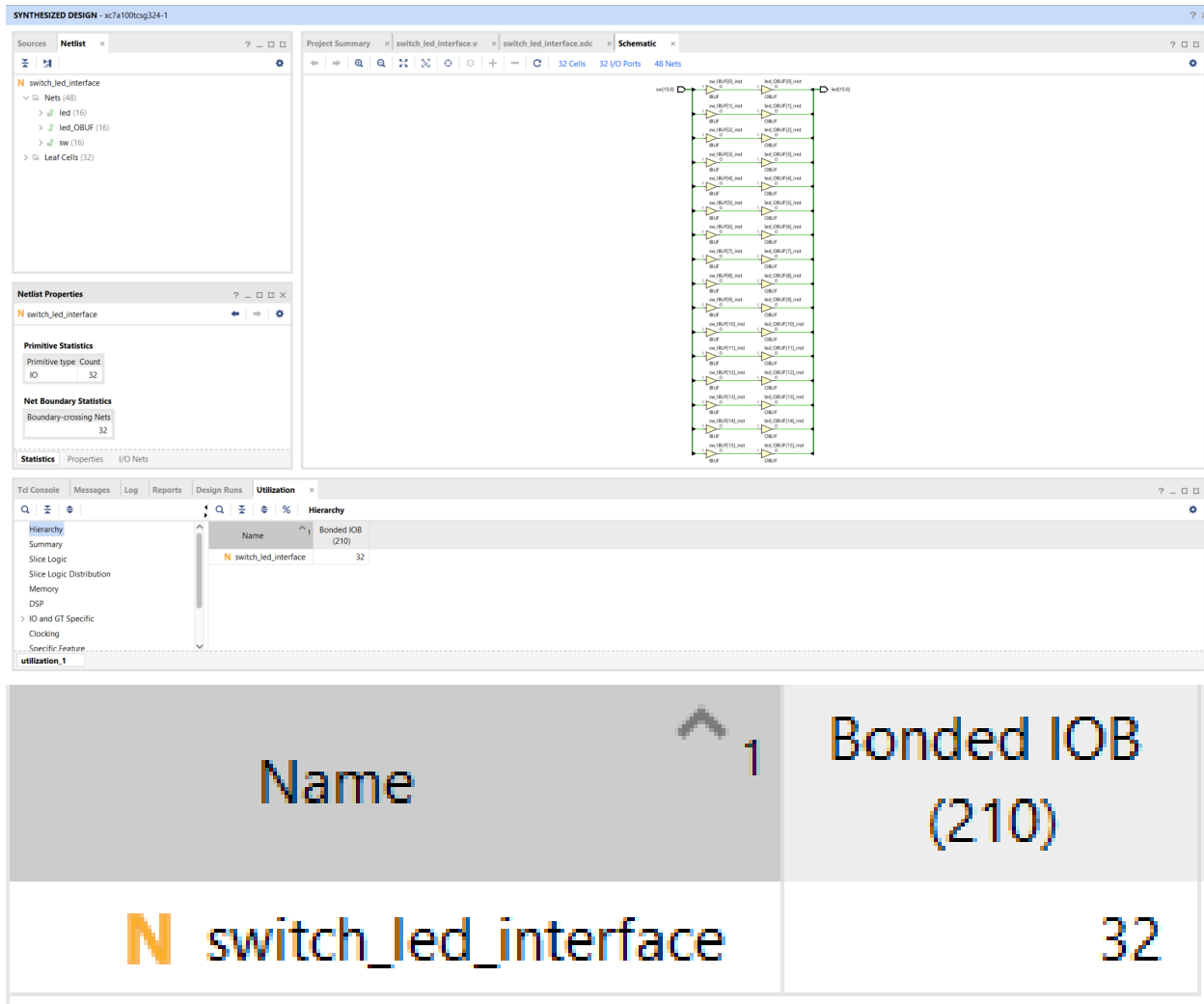
set_property PACKAGE_PIN V10 [get_ports {sw[15]}]
set_property IOSTANDARD LVCMOS33 [get_ports {sw[15]}]

## LEDs
set_property PACKAGE_PIN H17 [get_ports {led[0]}]
set_property IOSTANDARD LVCMOS33 [get_ports {led[0]}]

...

set_property PACKAGE_PIN V11 [get_ports {led[15]}]
set_property IOSTANDARD LVCMOS33 [get_ports {led[15]}]
```

Synthesis Screenshots:



Group Video:

<https://youtu.be/cNmZifSIaw0>

Reflections:

Ben Robles: This lab was a good intro to vivado and its workflow. With zero previous experience the basic project helped me connect all the gaps I had in my understanding from the lecture. While we didn't write any of the code ourselves, analyzing an example served its purpose and I feel prepared to work on a more involved lab.

Robert Stevenson: This was my first time using Vivado and programming an FPGA board so I appreciated that this lab gave me the experience that will be necessary for this lab and class. During this lab I learned how to write a test bench on verilog and how to use a constraints file as well as how to target and program an FPGA board using Vivado.